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Welcome page

On this website, you find a short tutorial that teaches you how to use the command line with bash, and how to connect to and work on an HPC.

You will first learn how to set up a terminal. Then the actual tutorial begins. In it, you take the role of a researcher who received data from a sequencing center, and who wants to start analysing this data on the command line. While you work, you keep a notebook in which you document what you did and why. The tutorial has two main parts:

1. In the first part, you will learn how to use bash on your own computer: you learn how to organize, inspect and extract information from your data.
2. In the second part, you will learn how to work with your data on an HPC: you learn how to connect to the HPC, copy your sequencing data to it, install software and run jobs that check and clean your sequencing data.

I wrote this tutorial for researchers at the Institute for Biodiversity and Ecosystem Dynamics (IBED) at the University of Amsterdam, who can get access to Crunchomics, the Genomics Compute Environment for SILS and IBED. If you don't have access to Crunchomics or another HPC that uses SLURM, you can still follow most of the tutorial on your own computer.

AI chat tools such as ChatGPT can write bash code for you, so why learn it yourself? Someone still has to run the code, read it and check what it did to your data, and that is what this tutorial teaches. At the end, the page Using LLMs describes how I suggest you use these tools once you know the basics.

For topics that this tutorial does not cover, have a look at IBED's bioinformatics guidance page.